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An Article Review on the Whitepaper "Business Analytics: New Concepts and Trends"

By

Peter-Paul D. Gumpal

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I. AN ARTICLE REVIEW ON THE WHITEPAPER "BUSINESS ANALYTICS: NEW CONCEPTS AND TRENDS"

Whitepaper/Journal: *Business Analytics: New Concepts and Trends*

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Source: *Management Journal of Sustainable Business and Management Solutions in Emerging Economies 25(2). An article from ResearchGate.*

Authors:

1. *Jasna Soldic Aleksic (Doctor of Philosophy) - University of Belgrade, Department of Mathematics and Statistics*
2. *Biljana Chroneos Krasavac (Doctor of Business Administration) - University of Belgrade, Department for Statistics and Mathematics*
3. *Ema Karamata (Master of Science, Doctor of Business Administration) - University of Belgrade, Business Management*

Reviewer:

1. *Peter-Paul D. Gumpal (Bachelor of Science major in Computer Science, 1999) - AMA University. 15 years of Applications and Database, Business Intelligence Development & Support experience, 10 years of Middle Management in FMCG and BPO companies.*

II. INTRODUCTION

This article review examines a research paper exploring Business Analytics (BA) and its role in driving organizational success. While the whitepaper highlights the importance of BA for gaining a competitive edge, the reviewer argues that a fundamental first step lies in digital transformation. Keeping financial and other company resource constraints in mind, this transformation should prioritize centralizing data sources, piecing together processes & functional groups, establishing efficient office workflows, and having a clear and consistent communication with stakeholders. Effective change management for digital transformation and data projects is crucial, ensuring staff, managers, and executives at all levels are committed to adopting best practices throughout the data journey – from scoping, data gathering, analysis and development, implementation, and cyclic continuous improvement. From the research's perspective, this collaborative approach will pave the way for a practical, sustainable, and trustworthy company IT universe – the cornerstone of successful BA initiatives.

This review aims to provide a clear and actionable perspective on BA in today's data-driven business environment by critically examining the whitepaper's strengths and potential areas for further exploration.

III. SUMMARY OF THE RESEARCH

The following are the key points in the whitepaper "*Business Analytics: New Concepts and Trends,* in which some of the Insights (in section IV) are contrasted.

1. The Significance of Business Analytics

The whitepaper emphasized Business Analytics (BA) as a critical tool for gaining a competitive advantage. It highlights how BA practices have evolved to address the ever-increasing data demands of modern businesses.

2. ICT is the driving force behind BA

The article acknowledges Information and Communication Technologies (ICT) as a major driver of BA advancements. It explores the role of technological infrastructure, including networking and data tools in enabling BA practices.

3. BA Techniques and employing Statistical Methods

The paper delves into various statistical methods used in BA. It discusses the three core analysis categories: Descriptive, Predictive, and Prescriptive, emphasizing their complementary nature in generating valuable insights and driving informed decision-making. Additionally, statistical tools like regression analysis, forecasting, and data mining techniques, etc. are mentioned, but were not explained in detail, or the author could have given a sample use case for these.

4. Democratization of Information

The paper addressed the evolving trend of making data analysis tools and insights accessible across all organizational levels such as Embedded BI and Streaming Analytics. Cloud technologies (for dashboarding and information dissemination) are mentioned as facilitators for anytime-anywhere access for employees, regardless of their role. The whitepaper also briefly touches on Open Initiatives, suggesting their potential economic impact at both macro and micro levels.

5. The Need for Data Expertise

The paper acknowledges the emergence of new data-related job titles like Data Engineers, Data Integrators, and Data Scientists, even C-levels such as Chief Data Officer. However, it lacked a detailed explanation of how these roles differ and collaborate within an organization. The paper could have placed a stronger emphasis on the growing need for talent to manage and analyze the ever-increasing volume of data collected by companies.

IV. INSIGHTS

The research paper was published in mid-2019. It offered valuable insights into the early influence of Information and Communication Technologies (ICT) on Business Analytics (BA) thereby the inception of BA periods that were coined as Analytics 1.0, Analytics 2.0, and Analytics 3.0. However, considering the rapid pace of technological advancements, particularly since 2019, significant knowledge gaps warrant further exploration. These advancements arguably contribute to Analytics version 4.0, and today, it has evolved beyond its "infancy" stage as the authors have suggested.

These insights based on the researcher's observations as a professional IT practitioner delve into critical areas not heavily emphasized in the paper, but which have become central to the BA domain:

1. The Data Surge from the New Normal

The COVID-19 pandemic significantly impacted global business practices, leading to a surge in data generation. The widespread adoption of online transactions, remote work collaboration, and virtual education has dramatically increased the volume and variety of data collected by organizations. This necessitates new approaches and tools (both hardware and software) within BA to effectively manage and analyze this ever-expanding data landscape. For instance, for the hardware, GPUs are now more prioritized for computational power to achieve faster processing (calculations) of handling blobs of data. In the software arena, API have grown organically to support Open Data initiatives, support FinTech-related transactions, and glue other Service Oriented Architectures that are now available (paid and free) on the Internet.

2. The AI and ML Revolution

The paper briefly mentions the Internet of Things (IoT), but the transformative impact of Artificial Intelligence (AI) and Machine Learning (ML) on BA practices remains largely unexplored, maybe because OpenAI's ChatGPT and Google's Bard were not yet commercialized back then. These technologies are revolutionizing BA by:

A. Automating Data Preparation - AI and ML algorithms can automate tedious data cleaning and preparation tasks, freeing up human analysts to focus on higher-level analysis and insights generation.

B. Deeper Insights - These powerful tools can uncover hidden patterns and relationships within complex data sets, leading to more profound and actionable insights for businesses.

C. Advanced Analytics - AI and ML enable more sophisticated forms of predictive and prescriptive analytics, allowing businesses to anticipate future trends and make data-driven decisions with greater confidence.

D. The rapid introduction of new trained models / LLM version in today's AI tools like OpenAI's ChatGPT from 3.5 to 4 and Google Bard to Gemini, further highlights the speedy evolving capabilities of AI as a tool for data prep and analyses. While these tools currently handle basic statistical calculations, their potential to generate Python and R code for Data Engineers and Data Scientists is particularly noteworthy.

3. The Expanding IoT Ecosystem and Mobile Data

The paper acknowledges the rise of IoT, but doesn't fully explore its impact. Since 2019, the proliferation of IoT devices has exploded, particularly in the health and security sectors: Smart wearables, connected home devices, and security cameras all generate data that can be integrated into BA practices for a more holistic view of customer behavior and operational efficiency.

Furthermore, the increasing affordability of smartphones and widespread social media adoption has contributed significantly to the data deluge. Understanding how to leverage data from these mobile sources is crucial for businesses in today's hyper-connected world.

4. The Ethical Imperative of BA

The paper doesn't delve into the ethical considerations surrounding BA practices. As data collection and analysis become more effective, concerns around privacy, and security become paramount. As an example here in the Philippines, personalized SMS messages is sent by a scammer to almost all Filipinos on a daily basis, and that scammer could have possibly acquired the data thru channels BA might have made possible. So it is essential to develop and implement ethical frameworks for data governance and responsible data-driven decision-making, not only for a private enterprise to adapt but also for our government's policy makers.

5. Three-pronged approach: The 3T

The paper acknowledges the multifaceted nature of achieving success with Business Analytics (BA). It outlines the need for robust tools and platforms, well-defined data strategies, and sound statistical methodologies. However, the paper could place greater emphasis on the crucial role of organizational commitment to digital transformation.

The reviewer emphasizes the importance of a three-pronged approach – the 3Ts – as the foundation for a successful BA implementation:

1. Tools & Technology - Implementing robust and up-to-date technological infrastructure is essential for managing, analyzing, and visualizing data effectively.
2. Talent - Building a skilled team of data analysts, data scientists, and other data professionals is crucial for extracting valuable insights from data.
3. Training - Providing ongoing training programs equips employees at all levels with the necessary skills to understand, interpret, and utilize data-driven insights for informed decision-making.

By prioritizing these three pillars – Tools, Talent, and Training – organizations can create a fertile environment for BA initiatives to flourish and ensure the technology becomes an organic part of the organizational culture. This "3Ts" framework acts as the glue that binds BA initiatives to the broader organizational strategy.

Aside from those insights mentioned, the reviewer might just be nit-picking but it lacks visuals, for example, “Infographics”. Information presented through infographics can be highly effective in simplifying complex concepts and enhancing reader comprehension. Charts, graphs, and other visual elements can make data trends and relationships readily apparent, allowing readers to grasp key points quickly and efficiently.

For example, the whitepaper’s mentione of “data tsunami”, the authors could have simply added this graph, this way the readers can appreciate the growth and projection being conveyed:

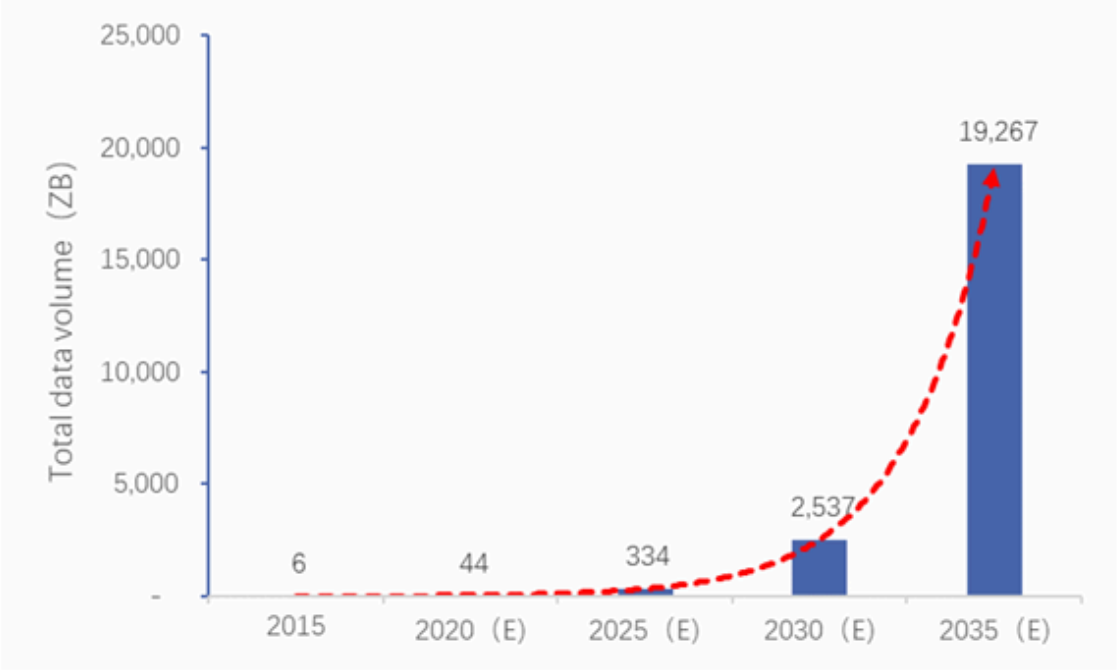
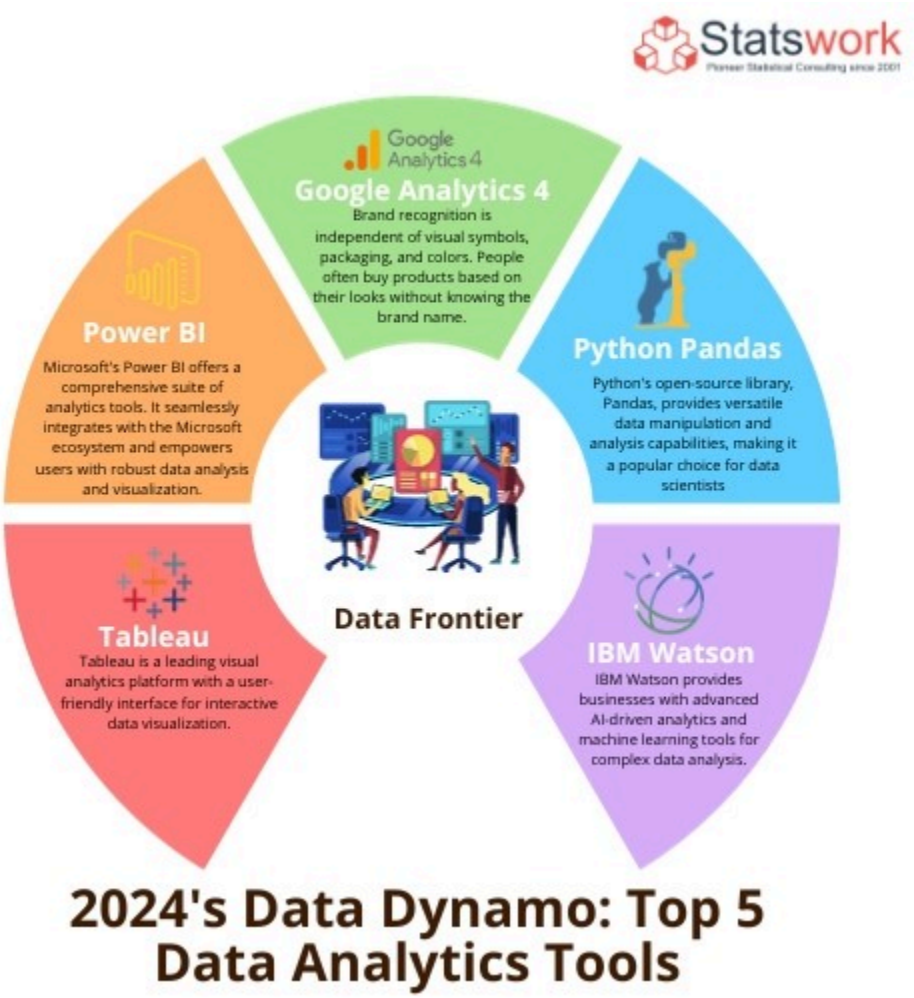


Fig. 1 The World’s Total Data Volume
Source: IDC, compiled by the Chinese Institute of Electronics, quoted from Ten Major Trends of the Global Digital Economy, 2018.

Source: *"South-South Cooperation in a Digital World. 2018 Annual Report in South-South Cooperation"*, published by United Nations Office for South-South Cooperation. (PDF from ResearchGate.com here)

Secondly, the authors could have mentioned real-world examples (i.e. Software. or Platforms that large organizations use for BI or BA). Providing concrete examples of Business Intelligence (BI) and BA platforms used by large organizations demonstrates the practical implementation of these concepts. This can be particularly beneficial for readers seeking to understand how BA translates from theory to real-world scenarios.

The author could have itemized some real IT tools. For example:



Source: “2024 Vision: Exploring The Impact And Evolution Of Advanced Analytics Tools”. Link here.

Lastly, case studies that could have made the understanding better for the reader. Including case studies showcasing successful BA implementations offers valuable lessons for readers. Learning how other organizations have leveraged BA to achieve specific goals can be highly insightful. These case studies can demonstrate the tangible benefits of BA practices across various industries and business contexts; the authors just referenced another study. The reviewer then tried to validate the authors' claim that the European Commission data initiatives will yield an economic value by 2020, but the reviewer wasn’t successful in finding an online document or another write-up that can back this claim now that we are in the year 2024. The paper wrote:

> *“Open data portals have powerful impact at both macro and micro-economic levels (Berends, Carrara, & Radu, 2017). The main economic benefits of open data are reflecting on: improved efficiency and quality of public administration, greater transparency of public services, new business models and innovative business*

services, as well as, various significant social contributions. It is estimated by Europe Commission that open data initiatives will contribute to the growth of EU data economy up to EUR 739 billion by 2020 that accounts for 4% of the EU GDP. The Economic Value of Open data for EU member states may be summarized as follows: direct market size of € 325 billion for the period 2016-2020, 100.000 jobs in Open Data in 2020 and € 1,7 billion cost savings for public sector in 2020 (Open Data Barometer, 2016, pp.3-4). The other study revealed the open data potential in the following areas – Consumer finance, Consumer products, Health care, Oil and gas, Education, Electricity and Transportation. It reveals that the economic potential of open data in these domains is approximately 3 - 5 trillion $ per year (Manyika, Chui, Farrell, Van Kuiken, Groves, & Almasi Doshi, 2013, p. 9)."

V. CONCLUSION

The researchers offered valuable insights into the early impact of ICT on BA. The paper accurately identified key trends like Big Data and data democratization that remain relevant even today.

However, the field of BA has evolved significantly since 2018. Advancements in Cloud technologies, the integration of Data Lakes, and the rise of Real-Time Analytics have further transformed how businesses collect, store, and analyze data (data pipeline)
From the perspective of Data Engineers, the role of Data Warehouses in managing Big Data and the importance of data governance in a democratized environment are crucial considerations. Additionally, the integration of AI, ML, and IoT data present exciting opportunities for BA, as these are now well blended together in a space that are easily accessible and almost-readily availble.

As the ICT landscape continues to evolve, so too will BA practices. By continuously re-examining the impact of these technologies and fostering responsible data-driven decision making, BA can remain a powerful tool for organizational success. But then again, the technology is only part of the equation.

A 2019 McKinsey Quarterly Report highlights the staggering cost of poor decision-making: a quarter of a billion dollars lost annually. (*"A survey we conducted recently with more than 1,200 managers across a range of global companies gave strong signs of growing levels of frustration with broken decision-making processes* [redacted, follow link for the full text] *The opportunity costs of this are staggering: about 530,000 days of managers' time potentially squandered each year for a typical Fortune 500 company, equivalent to some $250 million in wages annually"*, Source: McKinsey Quarterly, May 1, 2019). This problem often stems from fragmented data across an organization, where information resides in isolated silos and lacks consistency when used for strategic planning.

In conclusion, by recognizing both the continued relevance of established trends and the emergence of new technologies, organizations can leverage BA to its full potential and achieve a significant competitive advantage in the data-driven world.